Generative ConvNet Foundation Model With Sparse Modeling and Low-Frequency Reconstruction for Remote Sensing Image Interpretation

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***Abstract*— Foundation models offer a highly versatile and precise solution for intelligent interpretation of remote sensing images, thus greatly facilitating various remote sensing applications. Nevertheless, conventional remote sensing foundational models based on generative transformers neglect the consideration of multiscale features and frequency information, limiting their potential for dense prediction tasks in remote sensing scenarios. In this article, we make the first attempt to propose a generative convolutional neural network (ConvNet) foundation model tailored for remote sensing scenarios, which comprises two key components: First, a large dataset named GeoSense, containing approximately nine million diverse remote sensing images, is constructed to enhance the robustness and generalization of the foundation model during the pretraining phase. Second, a sparse modeling and low-frequency reconstruction (SMLFR) framework is designed for self-supervised representation learning of the ConvNet foundation model. Specifically, a sparse modeling strategy is proposed in masked image modeling (MIM), which allows ConvNet to process variable-length sequences by treating unmasked patches as voxels and sparsifying the encoder. In addition, a low-frequency reconstruction target is designed to guide the model's attention toward essential ground object features in remote sensing images, while mitigating unnecessary detail interference. To evaluate the general performance of our proposed foundation model, comprehensive experiments have been carried out on five datasets across three downstream tasks. Experimental results demonstrate that our method consistently achieves state-of-the-art performance across all the benchmark datasets and downstream tasks. The code and pretrained models will be available at https://github.com/HIT-SIRS/SMLFR.**

***Index Terms*— Convolutional neural network (ConvNet), foundation model, remote sensing, self-supervised pretraining.**

I. Introduction

In RECENT years, the field of remote sensing has been undergoing a transformative revolution due to the seamless integration of advances in information and communication technologies (ICTs). This has given rise to the emerging domain of intelligent remote sensing [1], which holds considerable potential in the realms of Earth observation data acquisition, optimization of resource management, and the augmentation of data interpretation capabilities. The success of intelligent remote sensing has benefited from advances in several disciplines, including satellite imaging, geographic information systems (GIS), image processing, machine learning, data analytics, statistical methodologies, cloud computing, and Earth sciences, enabling decision-makers to make judicious and sustainable choices pertaining to environmental monitoring [2], hazard assessment [3], urban planning [4], and precision agriculture [5]. Nonetheless, the distillation of pertinent knowledge from a myriad of remote sensing data sources poses a substantial hurdle. Conventional data mining methodologies frequently encounter impediments when endeavoring to unearth insightful revelations from the complex tapestry of such intricate information.

In contrast, deep learning has shown exceptional ability to process complex high-dimensional data, feature extraction, and learning high-quality image representations and has demonstrated commendable performance in various remote sensing scenarios. In spite of the advancements, deep-learning-based methodologies predominantly lean on supervised training, which necessitates large-scale, high-quality, and task-specific labeled datasets. However, the process of collecting and annotating such datasets requires significant time, resource allocation, and substantial financial investment, thereby setting a huge barrier for resource-constrained applications. Furthermore, the datasets acquired frequently exhibit constrained applicability, even analogous scenarios or applications within the remote sensing domains. This situation compels a need for repetitive iterations in both the data collection and model development phases. This iterative procedure not only contributes to the cumulative expenses in terms of time and finances but also acts as an impediment to the broader applicability, effectiveness, and expandability of deep learning frameworks in the context of remote sensing applications.

Many strategies have been proposed to address the aforementioned challenges, including transfer learning [6], few-shot learning [7], label-efficient learning [8], etc. Specifically, transfer learning, notably embraced in remote sensing applications and beyond, leverages deep learning models pretrained on large-scale image datasets such as ImageNet and Microsoft COCO. Few-shot learning emphasizes the training of models to rapidly adapt to novel tasks or domains with minimal annotations. This is achieved by harnessing prior knowledge or using meta-learning techniques. In contrast, label-efficient

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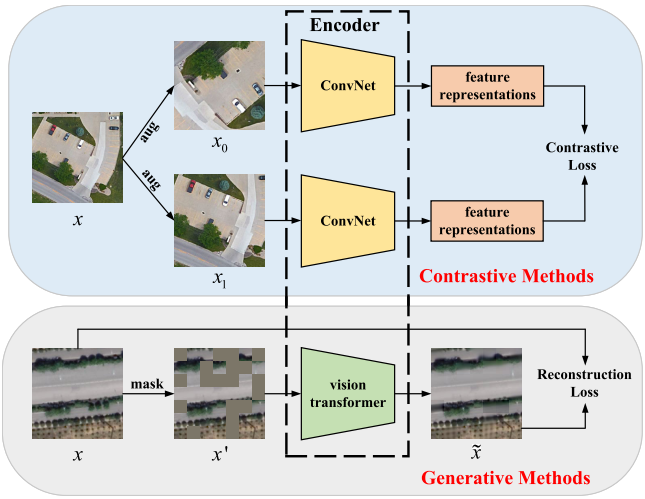


Fig. 1. Conceptual comparison between two mainstream remote sensing foundation models: ConvNets based on contrastive methods and ViT based on generative methods.

learning uses methods involving weak supervision or even the absence of supervision, aimed at alleviating the impact of demanding and time-consuming labeling complexities. Nevertheless, these approaches frequently undergo pretraining using data sourced from a single domain and modality, which limits their transferability and generalization.

Recently, foundation models [9] have demonstrated remarkable successes in the computer vision field. Foundation models are generally pretrained on large-scale datasets through self-supervised learning methods [10] and are capable of handling various downstream tasks with minimal fine-tuning labeled data. Furthermore, foundation models facilitate the cross-domain transfer and sharing of knowledge, diminishing the necessity for data exclusively tailored to specific tasks. Compared with the traditional artificial intelligence models, the advantages of foundation models can be summarized as follows.

1) *General Knowledge:* Through pretraining on extensive datasets, foundational models engender a form of "general intelligence" encompassing comprehensive knowledge within the domain of computer vision.
2) *Fine-Tuning Flexibility:* Foundational models excel in accommodating fine-tuning for specific tasks or datasets. It effectively circumvents the need for extensive model training from the ground up, thereby conserving computational resources and time investments.
3) *Data Efficiency:* Foundation models also showcase exceptional capabilities even when confronted with limited task-specific data, which are effective for real-world scenarios with annotation scarcity issues.

Recognizing the potential of foundational models, numerous researchers have commenced to explore the development of foundational models suitable for the field of remote sensing. The existing remote sensing foundational model construction methods include two mainstream methods: convolutional neural networks (ConvNets) based on contrastive methods [11] and vision transformers (ViTs) based on generative methods [12], as shown in Fig. 1. Specifically, contrastive methods aim to enhance the similarity between feature representations originating from two augmented views of the same image. The primary objective of generative methods is to facilitate the restoration of masked image pixels through prediction, thereby enabling coherent image reconstruction. Compared with contrastive methods that primarily focus on the central regions of images, generative approaches have the capability to learn features from each cropped patch, thereby acquiring representations better suited for transfer to downstream tasks. However, on one hand, since convolution operations cannot handle sequences of image patches, generative methods cannot be applied to ConvNets. Besides, taking intricate details in remote sensing images as reconstruction targets of generative methods would affect the performance of representation learning. On the other hand, the existing remote sensing datasets generally suffer from insufficient data quantity and diversity, which prevents foundation models from learning universal feature representations.

To tackle the above limitations, we propose a generative ConvNet foundation model for remote sensing image interpretations. Specifically, a sparse modeling and low-frequency reconstruction (SMLFR) self-supervised learning method is proposed for the pretraining of the ConvNet foundation model. We introduce sparse convolution to encode the irregular masked image and take the low-frequency input image as the reconstruction target. Besides, to improve the adaptability of the foundation model to different remote sensing scenarios, we construct a globally diverse remote sensing dataset named Geosense.

The main contribution of this article is threefold.

1) We propose the first generative ConvNet foundation model in the field of remote sensing, which achieves state-of-the-art performance on five datasets across three downstream tasks, demonstrating its effectiveness and generalizability.
2) An SMLFR self-supervised learning approach tailored for the ConvNet foundation model is proposed to enhance the quality of the learned representations by filtering the high-frequency intricate details from the reconstruction target.
3) To provide a diverse range of training samples for the pretraining of ConvNet foundational model, we have constructed the GeoSense remote sensing dataset. GeoSense comprises approximately nine million images sourced from various sensors, encompassing multiple continents and a wide array of scenes. This dataset offers the model a multitude of training samples, thereby enhancing the model's proficiency in recognizing and comprehending distinct objects and scenes, while simultaneously bolstering its robustness and generalization capabilities.

II. Related Work

A. Self-Supervised Learning

Self-supervised learning provides a paradigm for learning general feature representations from unlabeled data, which can be divided into three categories: predictive methods, contrastive methods, and generative methods.

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TABLE I
DETAILED STATISTICS OF THE GEOSENSE DATASET

Dataset	Scenes	Spatial Resolution	Size	Mean	Variance	Scale
GeoSense	urban, mountainous, rural, lakes, rivers, etc.	0.05 - 150m	224×224	[0.483, 0.454, 0.404]	[0.228, 0.223, 0.224]	8,916,233



Fig. 2. Some examples of the GeoSense dataset, where samples have (a) high intraclass diversity and (b) various imaging conditions. Besides, (c) spatial resolution between samples is highly variable and (d) each sample essentially contains multiple objects.

The predictive methods enable the model to acquire meaningful representations by engaging in specific pretext tasks that involve predicting certain attributes or contextual relationships in the data. Relative position [13] was designed to predict the relative position of random patch pairs. Similarly, Jigsaw [14] and rotation [15] were designed to predict the correct order of jigsaw puzzles and the rotation angle of the rotated image, respectively. Inpainting [16] was introduced to recover the missing patches of the whole image.

The contrastive methods focus on learning shared features among similar instances and distinguishing differences among dissimilar instances. It avoids the need to consider intricate details at the instance level, instead learning to discriminate data in an abstract semantic feature space. Momentum contrast (MoCo) [17] used momentum updates and query caching as key components, enabling efficient negative sampling using a dynamically updated cache of historical feature vectors. As the upgraded version, MoCo v2 [18] further optimized the sampling strategy, extended dictionary, and data augmentation. Simple framework for contrastive learning of visual representations (SimCLR) [19] proved the importance of hard positive sampling strategy. Unlike MoCo and SimCLR, bootstrap your own latent (BYOL) [20] reduced the reliance on large-scale negative samples by applying the mean square error (mse) loss. Simple siamese network (SimSiam) [21] achieved satisfactory results using Siamese networks and stopping gradient operators, eliminating the need for momentum encoders and large batches. Recently, several efforts have attempted to apply contrastive learning to ViT. Specifically, MoCov3 [22] extended the MoCo approach to ViT [23] by minimizing the distance between representations of two augmented views. MOBY [24] used MoCo v2 and BYOL simultaneously, creating a self-supervised framework based on the Swin [25] backbone. Self-distillation with no labels (DINO) [26] combined knowledge distillation, momentum encoder, and multicrop training to learn the local-to-global correspondence in the ViT. Moreover, dual contrastive derain-generative Adversarial network (DCD-GAN) [27] was proposed to bridge the domain gap between rainy and clean images using double contrast learning. Wu et al. [28] introduced an innovative approach called "learning from history" that enabled the adaptive generation of negative samples directly from the target model, facilitating task-agnostic and generic contrastive learning for image restoration. Practical contrastive learning framework for single-image super-resolution (PCL-SR) [29] achieved the adaptation of contrastive learning in single-image super-resolution by constructing positive and negative samples in frequency space.

The generative methods learn representations by reconstructing or generating input data, typically represented by masked autoencoders (MAE) in recent years. Bidirectional encoder representation from image transformers (BEiT) [30] was the first two-step autoencoder-based generative method for the computer vision field, mimicking bidirectional encoder (BERT) [31] to reconstruct visual tokens. Since image patches do not have preexisting tokens like words, BEiT trained an image tokenizer via a discrete variational autoencoder (dVAE) [32] before mask image modeling. To achieve end-to-end pretraining, MAE [33] was proposed to directly predict the masked patches from the unmasked components with mse loss. To save computation, only the unmasked patches were fed into the encoder of MAE. Unlike MAE, simple framework for masked image modeling (SimMIM) [34]'s encoder processed both visible tokens and masked tokens, and the decoder only applied a linear layer to regress pixel values.

B. Vision Foundation Model

With the advent of large-scale datasets and computational resources, ConvNets have become the dominant approach in the field of computer vision. Since AlexNet won the ImageNet Challenge in 2012, a number of deeper neural network architectures have been proposed, such as Visual Geometry Group (VGG), GoogleNet, residual networks (ResNet), ResNeXt, MobileNet, and EfficientNet. In addition to architectural innovations, advancements have also been made in convolutional operations, incorporating more sophisticated techniques such as depthwise convolution and deformable convolution. By leveraging the advanced principles underlying

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the transformers, modern ConvNets such as ConvNeXt [35], RepLKNet [36], sparse large Kernel network (SLaK) [37], HorNet [38], and InternImage [39] have demonstrated promising performance in visual tasks through the integration of large kernels and dynamic weighting mechanisms within the architecture designs.

In recent years, the development of fundamental models in the field of computer vision has focused on transformer-based architectures. ViT [23] stands as a prominent exemplar, having achieved remarkable success in visual tasks owing to its ability to capture global context and dynamic spatial aggregation. However, the global attention mechanisms used in ViT are hampered by their computationally and memory-intensive nature, which restricts their applicability in downstream tasks. To mitigate this issue, several approaches have been proposed. Pyramid ViT (PVT) [40] and linear transformer (Linformer) [41] adopted global attention mechanisms on downsampled key and value maps, while deformable attention transformer (DAT) [42] incorporated deformable attention to selectively sample information from value maps in a sparse manner. In addition, HaloNet [43] and Swin transformer [25] introduced local attention mechanisms and used techniques such as haloing and shift operations to facilitate information transfer among neighboring local regions.

C. Remote Sensing Foundation Model

Remote sensing foundation models address the need for scalable and general methods to handle rich and heterogeneous remote sensing data. Through self-supervised learning on massive and unlabeled remote sensing data, the foundation models can capture the inherent patterns, features, and representations present in remote sensing images, which can be applied to various downstream remote sensing tasks, including object detection, land cover classification, change detection, and map generation. Sun et al. [44] proposed the first remote sensing foundation model framework called RingMo, which achieved effective interpretation and analysis of remote sensing images through large-scale remote sensing datasets and masked image modeling (MIM) pretraining methods designed based on remote sensing data. Wang et al. [45] applied a plain ViT with approximately 100 million parameters as the foundation model and improved it for the characteristics of large-scale and different-oriented objects in remote sensing images. Specifically, a novel rotated varied-size window attention mechanism is introduced, which reduces the computational cost and memory footprint while improving object representation by extracting context from diverse windows. Cha et al. [46] proposed the first billion-level foundation model in the field of remote sensing and demonstrated that the performance and data efficiency of the foundation model increase with the increase in the number of parameters. Remote contrastive language-image pre-training (RemoteCLIP) [47] is the first multimodal visual-language foundational model for remote sensing, which aims to learn robust visual features with rich semantics, as well as aligned text embeddings.

Furthermore, some works have introduced self-supervised learning methods and foundational model architectures for specific remote sensing tasks. An adaptive self-supervised network for blind remote sensing image super-resolution was presented in [48], effectively reducing the domain gap between simulation and real-world data. The spatial and semantic consistency contrastive learning (SSCCL) [49] framework was designed to learn feature representations with spatial and semantic consistency for semantic segmentation. Xiao et al. [50] achieved temporal compensation in satellite video super-resolution by systematically modeling short-term and long-term temporal differences. Besides, an edge-enhancement network edge-enhanced (EEGAN) [51] was presented for robust satellite image super-resolution reconstruction, featuring an adversarial learning strategy resilient to noise. Ke et al. [52] proposed a coarse-to-fine adaption learning fusion network for remote sensing pansharpening. An enhanced spectral selection and extrapolation generative adversarial network (SS&E) [53] algorithm was proposed to use frequency-domain sorting physical model to constrain the self-supervised reconstruction of synthetic aperture radar (SAR) images. In addition, structured inheritance [54] learned new tasks while maintaining high performance on old tasks by introducing a pixel affinity structure loss and a representation consistency structure loss.

III. Proposed Method

A. GeoSense Dataset for Pretraining

We construct a globally diverse remote sensing dataset named GeoSense for pretraining fundamental models. The dataset is composed of publicly available remote sensing datasets [55], remote sensing image challenges [56], and other research projects and datasets. The detailed description of the dataset is presented in Table I. The images in the GeoSense dataset are acquired from sensors with different imaging parameters, including Sentinel-2, Gaofen, Landsat, QuickBird, etc. The diverse collection sources provide training samples with different attributes and features, enabling the model to capture variations and characteristics under different environmental conditions. In addition, the GeoSense dataset contains rich geographical and scene information. The images cover multiple continents of the world, including Asia, Europe, North America, South America, Africa, and Oceania, encompassing various scenes such as urban, rural, mountainous, lakes, and rivers. The diversity of samples in the GeoSense dataset helps the fundamental model learn a wide range of feature representations, enhancing its ability to recognize and understand various objects and scenes. Moreover, using diverse remote sensing data for pretraining, the fundamental model can better adapt to changes and interferences caused by factors such as weather, season, and lighting conditions, exhibiting improved robustness and generalization capabilities.

To facilitate pretraining, all the collected images are cropped to a size of 224×224 pixels, resulting in a large dataset of 8 916 233 training samples. These cropped image samples underwent consistent preprocessing, including normalization and standardization, to ensure color consistency across the samples. It is worth noting that we did not use the mean and variance from the ImageNet dataset, but instead computed them anew for all the samples in the GeoSense dataset. The GeoSense dataset exhibits a wide range of spatial resolutions, with the highest resolution being 0.05 m and the lowest resolution being 150 m. To the best of our knowledge, GeoSense

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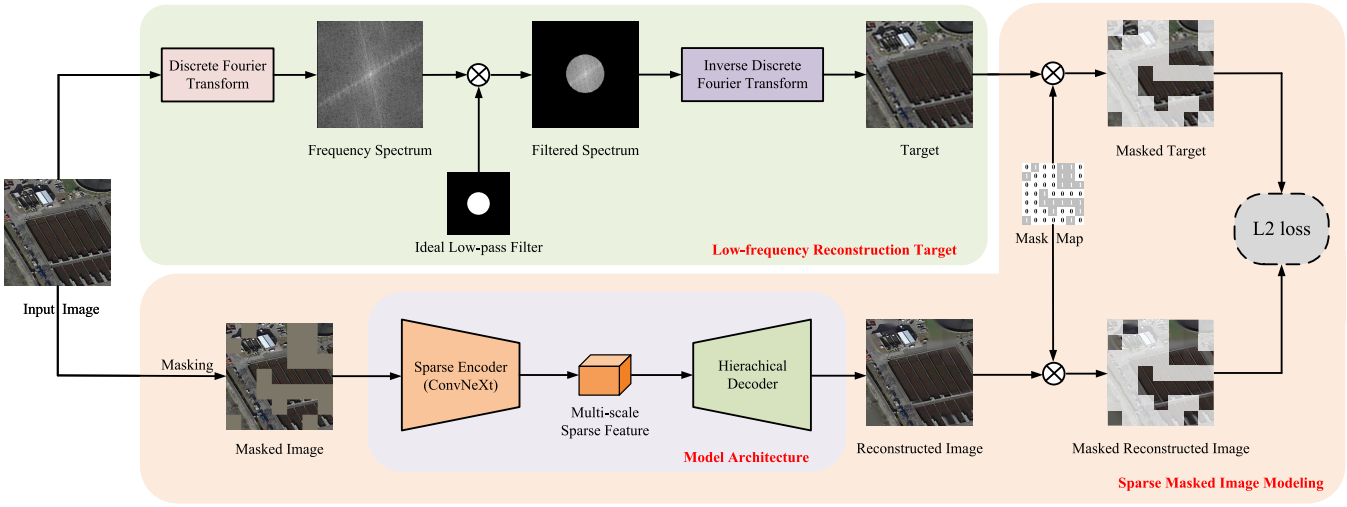


Fig. 3. Proposed SMLFR self-supervised learning framework is depicted in a bifurcated flowchart, featuring two parallel branches: one for building a reconstruction target and the other for MIM. In the above branch, the input image undergoes discrete Fourier transformation, transitioning it to the frequency domain. Circular low-pass filtering is then applied to eliminate high-frequency components, followed by inverse discrete Fourier transformation to reconstruct the image into a low-frequency target. Concurrently, the below branch conducts a masking operation on the input image and uses an encoder–decoder structure to restore the masked pixels. Subsequently, the SMLFR framework computes an $\mathcal{L}_2$ loss by evaluating the disparity in masked locations between the target and the reconstructed images generated by both the branches.

represents the largest scale remote sensing pretraining dataset to date. We select several data samples for presentation, shown in Fig. 2.

B. SMLFR Framework for Self-Supervised Learning

To effectively learn general universal representations from the GeoSense dataset, we propose an SMLFR self-supervised learning framework, which can be seen from Fig. 3. Below, we introduce the three components of SMLFR in detail: sparse modeling, model architecture, and low-frequency reconstruction.

1) Sparse Modeling: Previous MIM methods for ViT (e.g., MAE) split the image into fixed-size nonoverlapping patches and randomly mask them. The unmasked patches are then flattened and fed into the encoder. However, MAE cannot be implemented on ConvNet because ConvNet cannot handle variable-length sequence inputs.

Inspired by Woo et al. [57] and Tian et al. [58], after splitting the image into nonoverlapping patches with a given mask ratio, we gather all the unmasked patches and view them as a 2-D sparse array of pixels. In this way, the unmasked patches maintain the 2-D image structure and therefore can be processed by convolution operations. During the pretraining phase, we replace the standard convolution layers in the ConvNet encoder with submanifold sparse convolution layers to compute only at visible places. Besides, since dense images can be regarded as a special case of sparse images, the sparse encoder pretrained can be directly applied to the fine-tuning stage without any modification.

For the masking image, we apply the random masking strategy with a masking ratio of 0.6 and a mask patch size of 32. The masked image is fed to the sparse encoder to generate multiscale sparse feature maps, which are then processed by the hierarchical decoder to recover the masked regions of the original image. During self-supervised pretraining, the model is optimized with an $\mathcal{L}_2$ loss to minimize the normalized pixel error between the recovered image and the target on masked positions. After pretraining, the decoder is discarded and only the trained encoder is applied to downstream tasks. The reconstruction loss can be expressed as follows:

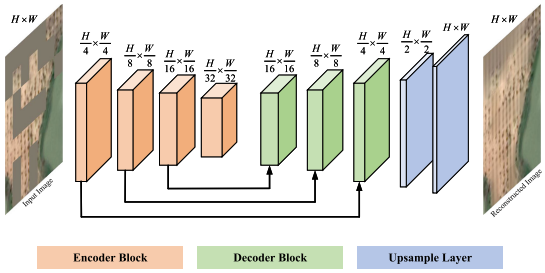


Fig. 4. Details of the model applied for SMLFR pretraining, which follows an encoder–decoder architecture. The encoder consists of four cascaded encoding blocks, while the decoder consists of three cascaded decoding blocks and two upsampling layers. Furthermore, we introduce skip connections between the corresponding encoder–decoder block pairs to decode multiscale sparse feature maps.

$$L_{\mathcal{L}_2} = \frac{1}{N}\sum_{i=1}^{N} \|x_i - y_i\|_2^2 \tag{1}$$

where N represents the number of masked pixels, and x_i and y_i represent the pixel values of the ith masked position in the reconstructed image and the target image, respectively.

2) Model Architecture: As shown in Fig. 3, the model is composed of an encoder and a decoder. In this study, ConvNeXt [35] is selected as the encoder, which has a hierarchical architecture and consists of four encoder blocks $\{E_1, E_2, E_3, E_4\}$, as depicted in Fig. 4. After the input image I with shape $H \times W$ is encoded through cascaded encoder blocks, four sparse feature maps of different scales $f_1 \in \mathbb{R}^{(H/4)\times(W/4)}$, $f_2 \in \mathbb{R}^{(H/8)\times(W/8)}$, $f_3 \in \mathbb{R}^{(H/16)\times(W/16)}$, and $f_4 \in \mathbb{R}^{(H/32)\times(W/32)}$ are generated.

For efficient computing, we introduce a lightweight decoder that contains three successive decoder blocks $\{D_1, D_2, D_3\}$ and two upsampling layers $\{U_1, U_2\}$, also shown in Fig. 4. Each decoder block consists of a transpose convolution, two

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3×3 convolutions, batch normalization, and rectified linear unit (ReLU) activation. Since sparse feature maps contain inactive empty positions, we fill them with mask tokens that obey the normal distribution with a mean of 0 and a standard deviation of 0.02 before decoding. This filling strategy can introduce some random noise to increase the robustness or learning ability of the model. Besides, we introduce skip connections in the asymmetric encoding–decoding structure to facilitate information transfer and feature fusion, thereby helping the model understand and capture details and global features in images.

3) Low-Frequency Reconstruction: The conventional MIM methods adopt the raw pixels as the reconstruction target, aiming to allow the model to perfectly reconstruct masked image patches. However, remote sensing images contain a large amount of noise and subtle texture details. These redundant information will increase the difficulty of the MIM task and cause the model to ignore the overall structure of the reconstructed target [59]. To allow the model to pay more attention to the overall shape, structure, and global characteristics of the ground objects in the remote sensing images, we filter out the high-frequency component of the input image and view the low-frequency image as the reconstruction target.

Specifically, we first apply the discrete Fourier transform (DFT) to convert the input image from the spatial domain to the frequency domain. For a certain channel I_c of the input image $I \in \mathbb{R}^{H\times W}$, the frequency representation of I_c can be expressed as follows:

$$\mathcal{F}(u, v) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} I_c(h, w)e^{-i2\pi\left(\frac{uh}{H}+\frac{vw}{W}\right)} \tag{2}$$

where $\mathcal{F}$ represents the converted frequency domain representation, (u, v) and (h, w) are the frequency spectrum and spatial coordinates, respectively, and i is the imaginary unit.

To preserve the low-frequency information while suppressing the high-frequency information in the input image, we introduce an ideal circular low-pass filter to extract the low-frequency components in the frequency representation

$$\mathcal{H}(u, v) = \begin{cases} 1, & \sqrt{\left((u-u_c)^2+(v-v_c)^2\right)} \le r \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $\mathcal{H}(\cdot)$ denotes the ideal circular low-pass filter, and u_c and v_c are the center coordinates of the frequency spectrum. $r \in [0, \min((H/2), (W/2))]$ is the cutoff frequency of the filter, which represents the threshold of the maximum frequency component allowed to pass. In this study, r is set to 40.

The low-frequency frequency representation can be represented as follows:

$$\mathcal{F}_{\text{low}}(u, v) = \mathcal{H}(u, v) \otimes \mathcal{F}(u, v) \tag{4}$$

where $\mathcal{F}_{\text{low}}(u, v)$ denotes the low-frequency frequency representation, and $\otimes$ is the elementwise multiplication.

Then we introduce the inverse DFT (IDFT) to convert the low-frequency frequency representation into spatial domain to generate the low-frequency reconstruction target $\mathcal{Y}$

$$\mathcal{Y}(x, y) = \frac{1}{UV}\sum_{u=0}^{U-1}\sum_{v=0}^{V-1} \mathcal{F}_{\text{low}}(u, v)e^{i2\pi\left(\frac{ux}{U}+\frac{vy}{V}\right)} \tag{5}$$

where U and V represent the horizontal and vertical dimensions of the frequency spectrum, respectively.

IV. Experiments

A. Foundation Model Pretraining

In the self-supervised pretraining phase, the foundation model is trained on our constructed GeoSense dataset. In this study, two ConvNeXt variants, ConvNeXt-B and ConvNeXt-L, were selected as the backbone architecture of the foundation model. The two variants only differ in the number of channels and the number of layers in each encoder block. After pretraining, the backbone network learns informative feature representations from a large-scale remote sensing dataset. It can be coupled with task-specific heads for transferring to various downstream tasks, such as object detection, semantic segmentation, and change detection, thereby reducing the reliance on extensive annotated data.

In our experimental setup, we implemented the pretraining process using the PyTorch framework. The pretraining process was conducted on eight Nvidia RTX A100 GPUs equipped with 80-GB memory, which allowed for efficient computation and optimization. To optimize the model's performance, we use the layer-wise adaptive moments optimizer for batching training (LAMB) optimizer and a cosine-annealing learning rate with peak value $= 0.0002 \times$ batchsize/256. For the data augmentation, only horizontal flipping and random cropping are used. We train the model for 400 epochs with a batch size of 1024. The model with the lowest loss during self-supervised pretraining is saved for subsequent downstream tasks.

Besides, to gain a qualitative understanding of the reconstruction task, we have conducted a visual analysis of several reconstruction results to illustrate the model's performance during the pretraining phase. These examples encompass a diverse range of geographic scenes and imaging conditions, providing a comprehensive demonstration of the model's effectiveness across various scenarios. As depicted in Fig. 5, it is evident that the model exhibits the capability to generate reasonable predictions for the masked regions by leveraging information from the unmasked adjacent patches. While it is acknowledged that the reconstructed pixels may appear slightly blurred, they predominantly retain essential feature information. This observation suggests that the model possesses a fundamental grasp of the contextual details of the overall scene.

B. Remote Sensing Object Detection

1) Dataset: In this section, we choose two benchmark datasets to evaluate the performance of pretrained foundation models: DOTA v1.0 [73] and DIOR-R [63]. The details of these two datasets are described in detail below.

a) DOTA v1.0: DOTA v1.0 is a large-scale dataset designed for aerial image object detection, comprising 2806 aerial images collected from various sensors and platforms. The images in the DOTA v1.0 dataset are obtained from Google Earth (GE), GF-2, and JL-1 satellites provided by the China Center for Resources Satellite Data and Application. Each image exhibits a diverse range of object instances, with

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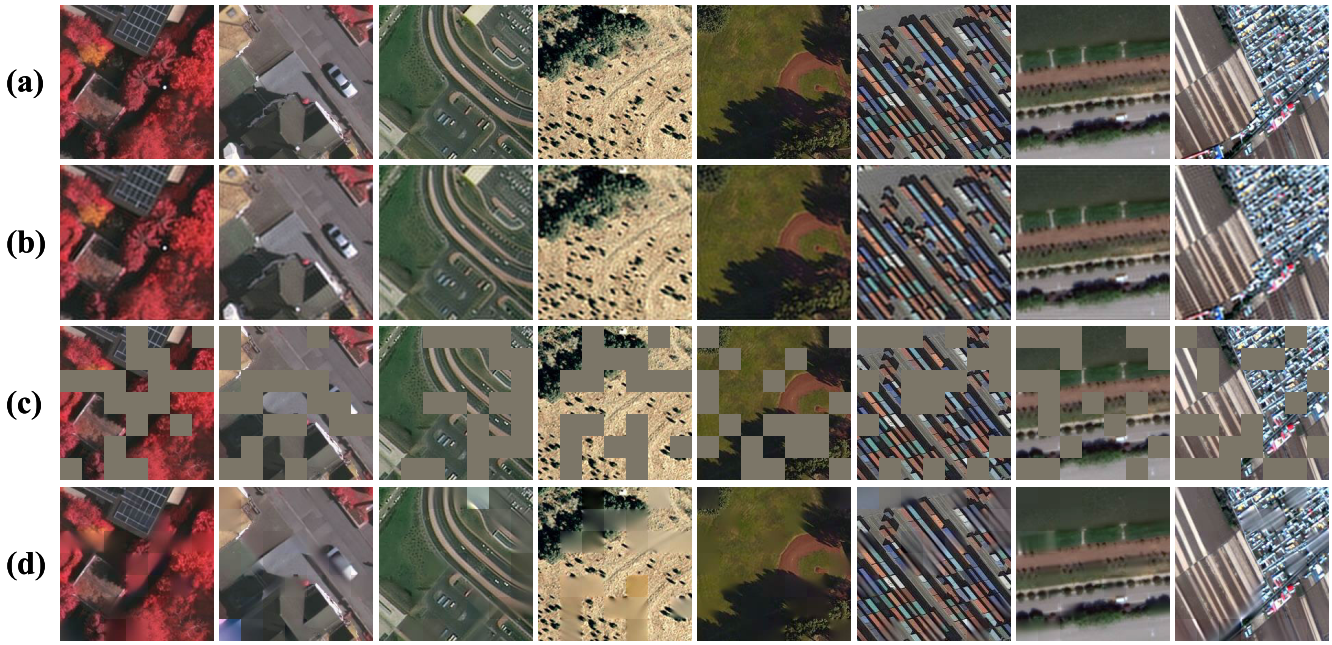


Fig. 5. Reconstructed examples of ConvNeXt-L pretrained by our proposed SMLFR. (a) Original images. (b) Low-frequency targets. (c) Masked images. (d) Reconstructed images.

TABLE II
COMPARATIVE RESULTS OF CLASSWISE AP AND MAP ON THE TEST SET OF THE DOTA-V1.0 DATASET

Method	AP per category (%)															mAP (%)
	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	
ROI Trans. [60]	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
R^3Det [61]	88.76	83.09	50.91	67.27	76.23	80.39	86.72	90.78	84.68	83.24	61.98	61.35	66.91	70.63	53.94	73.79
Gilding Vertex [62]	89.64	85.00	52.26	**77.34**	73.01	73.14	86.82	90.74	79.02	**86.81**	59.55	70.91	72.94	70.86	57.32	75.02
AOPG [63]	89.14	82.74	51.87	69.28	77.65	82.42	88.08	**90.89**	86.26	85.13	60.60	66.30	74.05	67.76	58.77	75.39
DODet [64]	89.61	83.10	51.43	72.02	79.16	81.99	87.71	**90.89**	86.53	84.56	62.21	65.38	71.98	70.79	61.93	75.89
S^2Aet [65]	88.70	81.41	54.28	69.75	78.04	80.54	88.04	90.69	84.75	86.22	65.03	65.81	76.16	73.37	58.86	76.11
ReDet [66]	88.79	82.64	53.97	74.00	78.13	84.06	88.04	**90.89**	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
R^3Det-KLD [67]	88.90	84.17	55.80	69.35	78.72	84.08	87.00	89.75	84.32	85.73	64.74	61.80	76.62	78.49	70.89	77.36
Oriented RepPoints [68]	89.11	82.32	56.71	74.95	**80.70**	83.73	87.67	90.81	87.11	85.85	63.60	68.60	75.95	73.54	63.76	77.63
Oriented R-CNN (ResNet-101) [69]	88.86	83.48	55.27	**76.92**	74.27	82.10	87.52	**90.90**	85.56	85.33	**65.51**	66.82	74.36	70.15	57.28	76.28
Oriented R-CNN (ViTAEv2-S) [69]	**89.66**	83.04	55.85	75.16	79.95	84.34	88.04	**90.90**	88.17	85.58	62.64	70.60	76.77	67.15	67.89	77.72
AO2-DETR [70]	89.27	84.97	56.67	74.89	78.87	82.73	87.35	90.50	84.68	85.41	61.97	69.96	74.68	72.39	**71.62**	77.73
Oriented R-CNN (ViT-B+RVSA) [45]	89.23	81.44	57.91	72.94	79.91	**85.08**	**88.23**	90.87	87.37	86.68	59.07	**73.62**	77.12	**78.70**	71.06	78.61
Oriented R-CNN (ViTAE-B+RVSA) [45]	89.48	83.44	57.78	71.85	80.33	**85.45**	**88.35**	90.88	**88.25**	86.59	62.17	72.09	77.32	**80.44**	69.35	78.92
Oriented R-CNN (SMLFR-ConvNeXt-B)	89.65	**86.88**	**58.72**	75.78	75.72	83.66	87.46	**90.90**	**88.19**	**87.39**	**66.93**	72.46	**77.62**	74.81	**71.20**	**79.15**
Oriented R-CNN (SMLFR-ConvNeXt-L)	**89.74**	**85.98**	**60.31**	73.96	**81.37**	84.42	88.08	**90.89**	85.96	86.66	64.67	**74.24**	**78.44**	74.34	70.99	**79.33**

* PL (plane), BD (baseball diamond), BR (bridge), GTF (ground track field), SV (small vehicle), LV (large vehicle), SH (ship), TC (tennis court), BC (baseball court), ST (storage tank), SBF (soccer ball field), RA (roundabout), HA (harbor), SP (swimming pool), HC (helicopter).

sizes varying from 800 × 800 to 4000 × 4000 pixels. The annotations in DOTA v1.0 are meticulously curated by aerial image interpretation experts, with a specific focus on 15 common object categories, including plane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, large vehicle, small vehicle, helicopter, roundabout, soccer ball field, and swimming pool. Inspired by the benchmark of oriented scene text detection, the annotations use oriented bounding boxes (OBBs) as they have been proven more suitable for accurately describing oriented instances in aerial images. In addition to the OBB, each instance is labeled with a category and a difficulty level indicating the extent to which it is challenging to detect. The DOTA v1.0 dataset contains a total of 2806 images and 188 282 instances, and it is divided into three parts: training set, validation set, and test set, which contain 1411, 458, and 937 images, respectively. As the DOTA v1.0 dataset is intended as a challenging benchmark, the labels for the test set are not publicly accessible, and thus, the inference results are uploaded to the online evaluation server[1] for performance evaluation.

b) DIOR-R: DIOR-R dataset is a publicly available and standard dataset for oriented object detection, which is an extension of the DIOR dataset [74] with relabeled oriented annotations. It comprises 23 463 images and 192 518

[1]http://bed4rs.net:8001/evaluation1/

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TABLE III
COMPARATIVE RESULTS OF CLASSWISE AP AND mAP ON THE TEST SET OF THE DIOR-R DATASET

Method	AP per category (%)																				mAP (%)
	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	
RetinaNet-O [71]	61.49	28.52	73.57	81.17	23.98	72.54	19.94	72.39	58.20	69.25	79.54	32.14	44.87	67.57	61.09	61.09	81.46	47.33	38.01	60.24	57.55
Faster R-CNN-O [72]	62.79	26.80	71.72	80.91	34.20	72.57	18.95	66.45	65.75	66.63	79.24	34.95	48.79	81.14	64.34	71.21	81.44	47.31	50.46	65.21	59.54
ROI-Trans. [60]	63.34	37.88	71.78	87.53	40.68	72.60	26.86	78.71	68.09	68.96	82.74	47.71	55.61	81.21	78.23	70.26	81.61	54.86	43.27	65.52	63.87
Gilding Vertex [62]	65.35	28.87	74.96	81.33	33.88	74.31	19.58	70.72	64.70	72.30	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06
AOPG [63]	62.39	37.79	71.62	87.63	40.90	72.47	31.08	65.42	77.99	73.20	81.94	42.32	54.45	81.17	72.69	71.31	81.49	60.04	**52.38**	**69.99**	64.41
DODet [64]	63.40	43.35	72.11	81.32	43.12	72.59	33.32	78.77	70.84	74.15	75.47	48.00	59.31	85.41	74.04	**71.56**	81.52	55.47	**51.86**	66.40	65.10
Oriented R-CNN (ViT-B+RVSA) [45]	80.92	49.88	81.05	88.52	51.52	80.17	**37.87**	75.96	88.83	78.46	84.01	46.53	64.18	81.21	84.04	71.34	89.99	65.41	50.53	66.49	70.85
Oriented R-CNN (ViTAE-B+RVSA) [45]	81.20	**54.71**	81.12	88.13	**51.83**	79.93	36.79	76.06	**89.23**	78.30	84.46	47.29	**65.01**	81.19	82.17	70.69	90.03	**66.75**	50.73	65.40	71.05
FROBB+RT (ViT-B12×1) [46]	72.25	53.86	80.55	81.49	45.15	79.96	31.56	71.44	85.42	78.67	83.72	47.57	59.55	81.26	84.87	71.28	81.51	64.73	49.43	66.05	68.52
FROBB+RT (ViT-L12×4) [46]	81.20	**60.47**	81.01	**89.97**	**51.76**	80.46	**39.98**	78.75	89.12	**78.77**	84.06	**53.85**	60.93	81.24	84.40	**71.77**	**90.15**	**66.22**	51.46	**66.59**	**72.11**
Oriented R-CNN(SMLFR-ConvNeXt-B)	**81.57**	51.55	**81.20**	89.58	50.40	**81.10**	36.38	**79.85**	88.12	78.73	**84.98**	51.47	64.02	**89.46**	**86.54**	62.89	90.04	65.68	50.05	66.40	71.50
Oriented R-CNN(SMLFR-ConvNeXt-L)	**81.61**	52.69	**89.06**	**89.70**	51.11	**81.33**	36.52	**80.48**	**89.36**	**79.17**	**85.69**	**53.81**	**64.22**	**89.77**	**86.62**	63.05	**90.28**	65.64	50.79	65.79	**72.33**

* APL (airplane), APO (airport), BF (baseballfield), BC (basketballcourt), BR (bridge), CH (chimney), DAM (dam), ETS (expressway-toll-station), ESA (expressway-service-area), GF (golffield), GTF (groundtrackfield), HA (harbor), OP (overpass), SH (ship), STA (stadium), STO (storagetank), TC (tenniscourt), TS (trainstation), VE (vehicle), WM (windmill).

object instances, providing a large-scale dataset for research purposes. The dataset focuses on 20 common object categories found in aerial images, including airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service area, expressway toll station, golf field, ground track field, harbor, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill. All the images in the dataset have a fixed size of 800×800 pixels, with varying spatial resolutions ranging from 0.5 to 30 m. The dataset is divided into three sets: training, validation, and testing. The training and validation sets consist of 11 725 images and 68 073 instances, while the testing set includes 11 738 images and 124 445 instances. Since the DIOR-R test set provides annotations corresponding to instances, we evaluate our model locally.

2) Implementation Details and Experimental Settings: Since the spatial dimensions of images in the DOTA v1.0 dataset are irregular, we crop the original images into 1024×1024 patches with an overlap of 200 pixels. In contrast, the DIOR-R dataset maintains a consistent image size of 800×800 pixels, eliminating the need for any cropping operations. We adopt the same hyperparameter configuration on the DOTA v1.0 and DIOR-R datasets. During the training phase, the training and validation sets are mixed together, and only random horizontal, vertical, and diagonal flips are applied as data augmentation. In the inference phase, we do not use any data augmentation. Average precision (AP) for each category and mean AP (mAP) are used as metrics to evaluate performance.

For a fair comparison, we select the most commonly used Oriented R-CNN [69] as the detection head, and pretrained ConvNeXt with feature pyramid network (FPN) [75] is viewed as the backbone network. All the experiments were implemented based on the MMRotate framework and eight NVIDIA A100 GPUs. We adopt the AdamW optimizer with a learning rate of 0.0001 and the weight decay set to 0.05. In adherence to the prevailing norm, the batch size is set to 16 and the training schedule is 12 epochs. Notably, we incorporate a learning rate reduction strategy that reduces the learning rate by a factor of 10 at the 8th and 11th epochs, promoting convergence and fine-tuning as training progresses. In addition, we use a stagewise learning rate decay strategy specifically targeted at the first 12 layers, with a decay rate of 0.9. For the task of object detection, we used two distinct loss functions: cross-entropy loss for class prediction and smooth L_1 loss for bounding box localization. Both the loss functions carry equal weight, emphasizing the same importance of both the aspects.

3) Experimental Results: Tables II and III show the comparative experimental results of remote sensing object detection on DOTA v1.0 and DIOR-R, respectively. Optimal and suboptimal performance in each metric are marked by **red** and **blue**.

On the DOTA v1.0 dataset, we conduct a comparative analysis of two variants of our proposed foundation models, SMLFR-ConvNeXt-B and SMLFR-ConvNeXt-L, against several state-of-the-art methods. The results of the comparisons are presented in Table II. It is worth noting that SMLFR-ConvNeXt-B demonstrates remarkable performance across various categories, exceling particularly in "baseball diamond," "tennis court," "storage tank," and "soccer ball field." This performance can be attributed to the models' ability to effectively capture the intricate features and spatial relationships characteristic of these object classes. Notably, in terms of mAP metric, SMLFR-ConvNeXt-B surpasses the previously leading method, vision transformer advanced by exploring intrinsic IB from convolutions+rotated varied-size attention (VITAE-B+RVSA) [45], by a margin of 0.23 percentage points, even though both the models operate with comparable parameter levels. Furthermore, when we further increase

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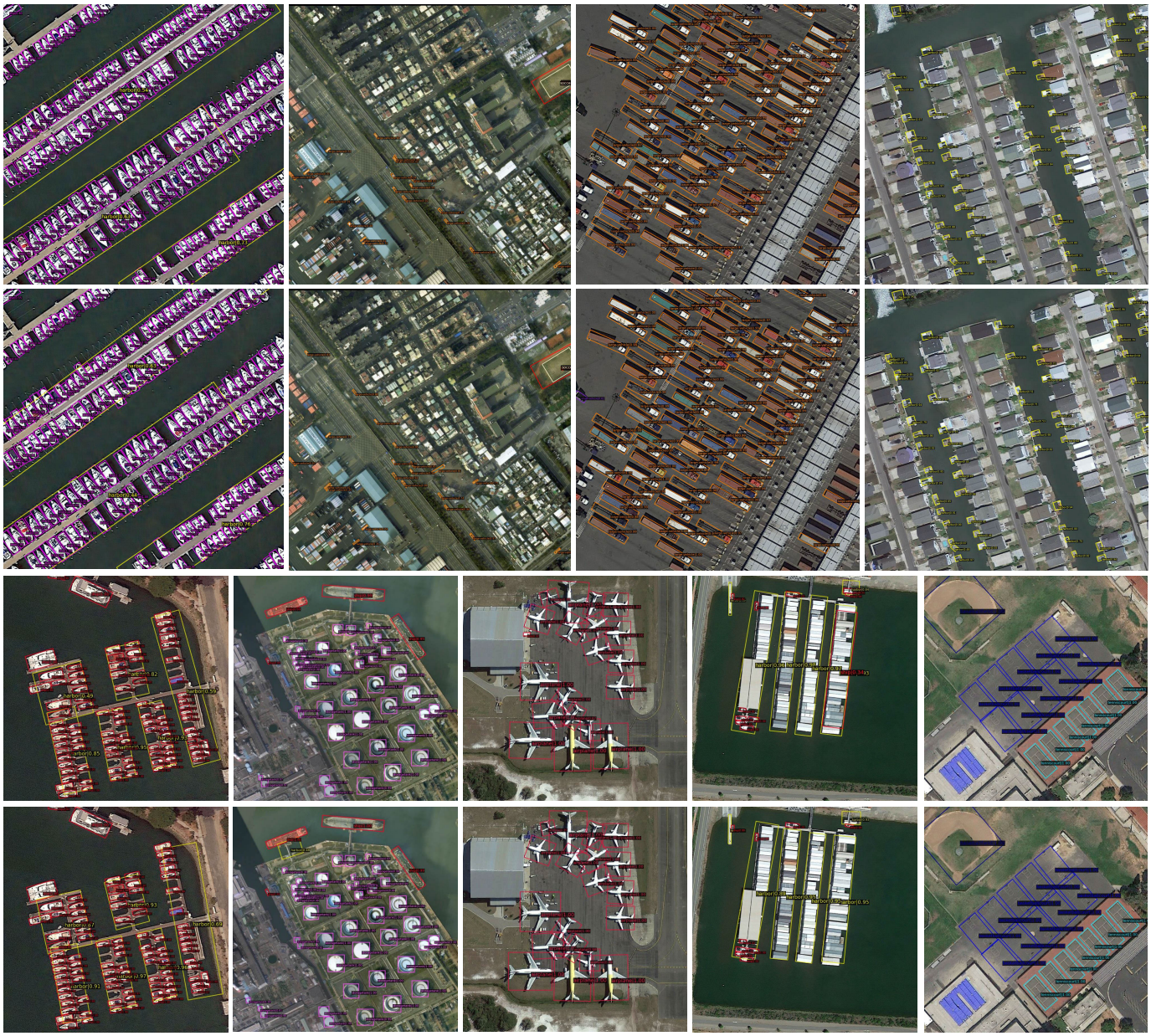


Fig. 6. Visualization results of the proposed model on the object detection task. The first two rows are the results of SMLFR-ConvNeXt-B and SMLFR-ConvNeXt-L on the DOTA v1.0 dataset, and the last two rows are the results of SMLFR-ConvNeXt-B and SMLFR-ConvNeXt-L on the DIOR-R dataset.

the number of parameters and model complexity, SMLFR-ConvNeXt-L showcases superior object detection performance compared with SMLFR-ConvNeXt-B. SMLFR-ConvNeXt-L demonstrates its capabilities in handling complex object categories, achieving notable improvements in categories such as "bridge," "large vehicle," and "harbor." This highlights the models' adaptability and capacity to excel in scenarios that demand a higher level of feature representation.

We further compare our models with several state-of-the-art methods on the DIOR-R dataset, and the comparative results are listed in Table III. Under the condition of similar parameter magnitude, SMLFR-ConvNeXt-B exceeds the previous best method ViTAE-B+RVSA [45] by 0.45% mAP. Similarly, SMLFR-ConvNeXt-L exceeds ViT-L12 $\times$ 4 [46] by 0.22% mAP. Besides, our models perform well in categories such as "airplane," "baseballfield," "groundtrackfiled," "ship," and "stadium," which often contain large-scale targets and various complex structures. This highlights the advantages of our method in handling complex scenes and large-scale target detection tasks. We also provide some visualization results in Fig. 6, and it can be seen that two variants of our proposed foundation models can both accurately identify and locate targets with high confidence under different scene and object distribution conditions.

C. Remote Sensing Semantic Segmentation

1) Dataset: In this section, the Potsdam[2] and LoveDA [56] datasets were selected as the benchmark datasets for comparison experiments to evaluate the effectiveness and generalization of the proposed models.

[2]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

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TABLE IV
COMPARATIVE RESULTS OF CLASSWISE F1 SCORE, MEAN F1 SCORE, AND OA ON THE TEST SET OF THE POTSDAM DATASET

Method	F1 score per category (%)					mF1 (%)	OA (%)
	Imper. surf.	Building	Low veg.	Tree	Car		
FCN [76]	88.61	93.29	83.29	79.83	93.02	87.61	85.59
DANet [77]	91.61	96.44	86.11	88.04	83.54	89.14	89.72
PSPNet [78]	91.61	96.30	86.41	86.84	91.38	90.51	89.45
DeeplabV3+ [79]	92.35	96.77	85.22	86.79	93.58	90.94	89.74
UperNet [80]	93.14	96.75	86.30	86.13	91.56	90.78	91.26
S-RA-FCN [81]	91.33	94.70	86.81	83.47	**94.52**	90.17	88.59
UZ_1 [82]	89.30	95.40	81.80	80.50	86.50	86.70	85.80
UFMG_4 [83]	90.80	95.60	84.40	84.30	92.40	89.50	87.90
Multi-filter CNN [84]	90.94	96.98	76.32	73.37	88.55	85.23	90.65
HMANet [85]	92.38	96.08	86.93	**88.21**	**95.44**	**91.81**	90.46
LANet [86]	93.05	**97.19**	**87.30**	**88.04**	94.19	**91.95**	90.84
UperNet-Seco [87]	91.21	94.92	85.12	84.89	89.02	89.03	89.64
UperNet(RSP-ViTAEv2-S) [88]	93.05	96.62	86.62	85.89	91.01	90.64	91.21
UperNet(ViT-B+RVSA) [45]	92.52	96.15	86.27	85.57	90.69	90.24	90.77
UperNet(ViTAE-B+RVSA) [45]	92.97	96.70	86.68	85.92	90.93	90.64	91.22
UperNet(RingMo) [44]	**93.63**	**97.13**	87.08	86.36	92.16	91.27	**91.74**
UperNet(SMLFR-ConvNeXt-B)	93.41	96.82	87.28	86.93	91.89	91.27	91.72
UperNet(SMLFR-ConvNeXt-L)	**93.47**	96.95	**87.35**	87.11	92.30	91.44	**91.82**

a) Potsdam: The Potsdam dataset published by the international society for photogrammetry and remote sensing (ISPRS) consists of 38 high-definition true orthophoto (TOP) tiles of the fixed size of 6000×6000 with a spatial resolution of 0.5 m. In all, 24 tiles are selected for training, while the remaining 14 tiles are reserved for testing. The dataset is manually annotated with six categories: impervious surfaces, building, low vegetation, tree, car, and clutter. F1 score and overall accuracy (OA) are served as the evaluation metrics, with the exclusion of category clutter. Since the annotations of the dataset are fully accessible, the test set is applied for performance evaluation.

b) LoveDA: The LoveDA dataset contains 5987 images with a spatial size of 1024×1024 pixels, with 2522 images for training, 1669 images for validation, and 1796 images for testing. Each image is composed of red, blue, and green spectral bands. The dataset is divided into two distinct parts: urban and rural, representing different environmental settings. It is meticulously annotated with seven categories for comprehensive semantic segmentation: background, buildings, roads, water, barren land, forests, and agriculture. However, the background category is excluded during training and evaluation. To better highlight the overall performance, urban and rural parts are integrated for training and testing. F1 score and intersection over union (IoU) metrics are used for performance estimating. Since the annotations of the test set are not published, we evaluate the models by submitting the inference results to the online evaluation server.[3]

2) Implementation Details and Experimental Settings: For the images in the Potsdam dataset, we use a sliding window strategy to crop them into patches of size 512×512, with a stride of 256×256 pixels. In the case of the LoveDA dataset, we perform random cropping of 1024×1024 images into 512×512 patches. During the training phase, we apply basic data augmentation techniques to both the datasets, which include random scaling within a ratio range of 0.5–2.0, random flipping, and photometric distortion. It is important to note that for the LoveDA dataset, we conduct inference on test images at their original spatial size of 1024×1024. To align with the commonly used segmentation evaluation criteria, we report F1 scores for individual classes, the mean F1 score (mF1), and OA metrics for the Potsdam dataset. In addition, during model training and testing, we exclude the clutter class. For the LoveDA dataset, we provide F1 scores for each class and the mean of the IoU (mIoU) as evaluation metrics.

For a fair comparison, UperNet [80] is selected as the semantic segmentation head. We conduct all the experiments on the MMSegmentation framework and eight NVIDIA A100 GPUs. The number of samples on each GPU is two. We use the AdamW optimizer for training the models, setting the base learning rate to $1e^{-4}$ and applying a weight decay of 0.05. Similar to our setting in object detection, we implement a stagewise learning rate decay strategy, which is primarily directed at the initial 12 layers of the network with a decay rate of 0.9. In addition, our training process consists of 160 000 iterations, and we incorporate a linear warm-up phase within the initial 1500 iterations, adjusting the warm-up ratio to $1e^{-6}$. To optimize our model for semantic segmentation task, we adopted the cross-entropy loss function, which measures the dissimilarity between predicted pixel-wise class probabilities and the corresponding ground-truth labels.

3) Experimental Results: Tables IV and V present a comparative analysis of experimental outcomes for remote sensing semantic segmentation task conducted on the Potsdam and

[3]https://codalab.lisn.upsaclay.fr/competitions/421#learn_the_details-overview

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TABLE V
COMPARATIVE RESULTS OF CLASSWISE F1 SCORE AND MIOU ON THE TEST SET OF THE LOVEDA DATASET

Method	F1 score per category (%)							mIoU (%)
	Background	Building	Road	Water	Barren	Forest	Agriculture	
FCN8S [76]	42.60	49.51	48.05	73.09	11.84	43.49	58.30	46.69
DeepLabV3+ [79]	42.97	50.88	52.02	74.36	10.40	44.21	58.53	47.62
PSPNet [78]	44.43	52.16	53.51	76.54	9.76	44.12	57.91	48.34
PAN [89]	43.04	51.34	50.93	74.77	10.03	42.19	57.65	47.13
UNet [90]	43.06	52.74	52.78	73.08	10.33	43.05	59.87	47.84
UNet++ [91]	42.85	52.58	52.82	74.51	11.42	44.42	58.80	48.20
Semantic-FPN [92]	42.93	51.53	53.43	74.67	11.21	44.62	58.68	48.15
LinkNet [93]	43.61	52.07	52.53	76.85	12.16	45.05	57.25	48.50
FarSeg [94]	43.09	51.48	53.85	76.61	9.78	43.33	58.90	48.15
FactSeg [95]	42.60	53.63	52.79	76.94	16.20	42.92	57.50	48.94
HRNet [96]	44.61	55.34	**57.42**	73.96	11.07	45.25	60.88	49.79
UperNet(ViT-B+RVSA) [45]	46.62	57.89	56.93	78.74	15.88	46.34	61.25	51.95
UperNet(ViTAE-B+RVSA) [45]	**46.69**	58.14	57.12	79.66	16.55	46.46	62.44	52.44
UperNet(ViT-B12×1) [46]	45.69	**58.75**	56.70	76.56	10.56	**48.52**	62.16	51.28
UperNet(ViT-L12×4) [46]	46.17	**60.56**	57.26	76.95	16.05	**47.50**	62.17	52.38
UperNet(SMLFR-ConvNeXt-B)	46.19	57.57	56.22	**80.39**	**16.94**	47.21	**63.59**	**52.59**
UperNet(SMLFR-ConvNeXt-L)	**47.05**	58.26	**57.37**	**80.64**	**17.64**	46.88	**63.34**	**53.03**

LoveDA datasets, respectively. Tables IV and V highlight the optimal and suboptimal performance for each evaluated metric using **red** and **blue**.

In Table IV, we present a comprehensive comparative analysis of various methods applied to the Potsdam dataset. Our proposed models excel in terms of classwise F1 scores, consistently outperforming the majority of other methods across all the categories. Notably, for the "Low veg." category, our SMLFR-ConvNeXt-L model achieves an impressive F1 score of 87.35%, ranking among the top performance in Table IV. This underscores the effectiveness of our approach in accurately delineating vegetation structures within the Potsdam dataset. In addition, our methods exhibit remarkable performance in both mF1 and OA, underscoring their effectiveness in addressing semantic segmentation challenges on the Potsdam dataset.

Table V presents the segmentation results achieved by our models in comparison to other state-of-the-art methods on the LoveDA dataset. It can be seen that our proposed models, SMLFR-ConvNeXt-B and SMLFR-ConvNeXt-L, showcase remarkable robustness across a variety of semantic categories. Particularly, our models excel in the "Water" and "Agriculture" categories, showcasing promising performance. However, they encounter challenges in precisely delineating areas labeled as "Background" and "Barren." This suggests that distinguishing regions with similar visual characteristics remains a formidable task. Despite these challenges, our methods achieve commendable mIoU values, registering 52.59% for SMLFR-ConvNeXt-B and 53.03% for SMLFR-ConvNeXt-L, which surpass the performance of the state-of-the-art approaches. This underscores our methods' capacity to yield a well-rounded segmentation performance encompassing all the categories. The visualization examples in Fig. 7 further highlight the adaptability and accuracy of the proposed models in complex and diverse remote sensing scenarios.

D. Remote Sensing Change Detection

1) Dataset: In this section, we conduct experiments on the LEVIR-CD [97] dataset to verify the effectiveness of our models.

a) LEVIR-CD: The LEVIR-CD dataset is a public and large-scale building change detection dataset, which consists of 637 pairs of images of 1024×1024 pixels with a high spatial resolution of 0.5 m. In all, 445 image pairs are for training, 64 image pairs for validation, and the rest 128 image pairs for testing. All the images in the dataset were collected using the GE API within the state of Texas, USA, spanning various capture dates ranging from 2002 to 2018. The most changes in the dataset are from man-made structures, including 31 333 independent building change entities.

2) Implementation Details and Experimental Settings: Images from the LEVIR-CD dataset undergo cropping into nonoverlapping patches with dimensions of 256 × 256 pixels. Our data augmentation strategy includes standard techniques such as random rotation, cropping, and flipping, which are applied to both the images within each training image pair. For evaluating the performance, we primarily use the F1 score, precision, and recall metrics specifically pertaining to the change category.

The change detection model used in our experiments is based on bitemporal image transformer (BIT) [107], with our pretrained SMLFR-ConvNeXt serving as the backbone. We conduct all the experiments on eight NVIDIA A100 GPUs using the Open-CD framework. We train the model for 2000 iterations, and the batch size is 16. Furthermore, it is worth noting that the hyperparameter

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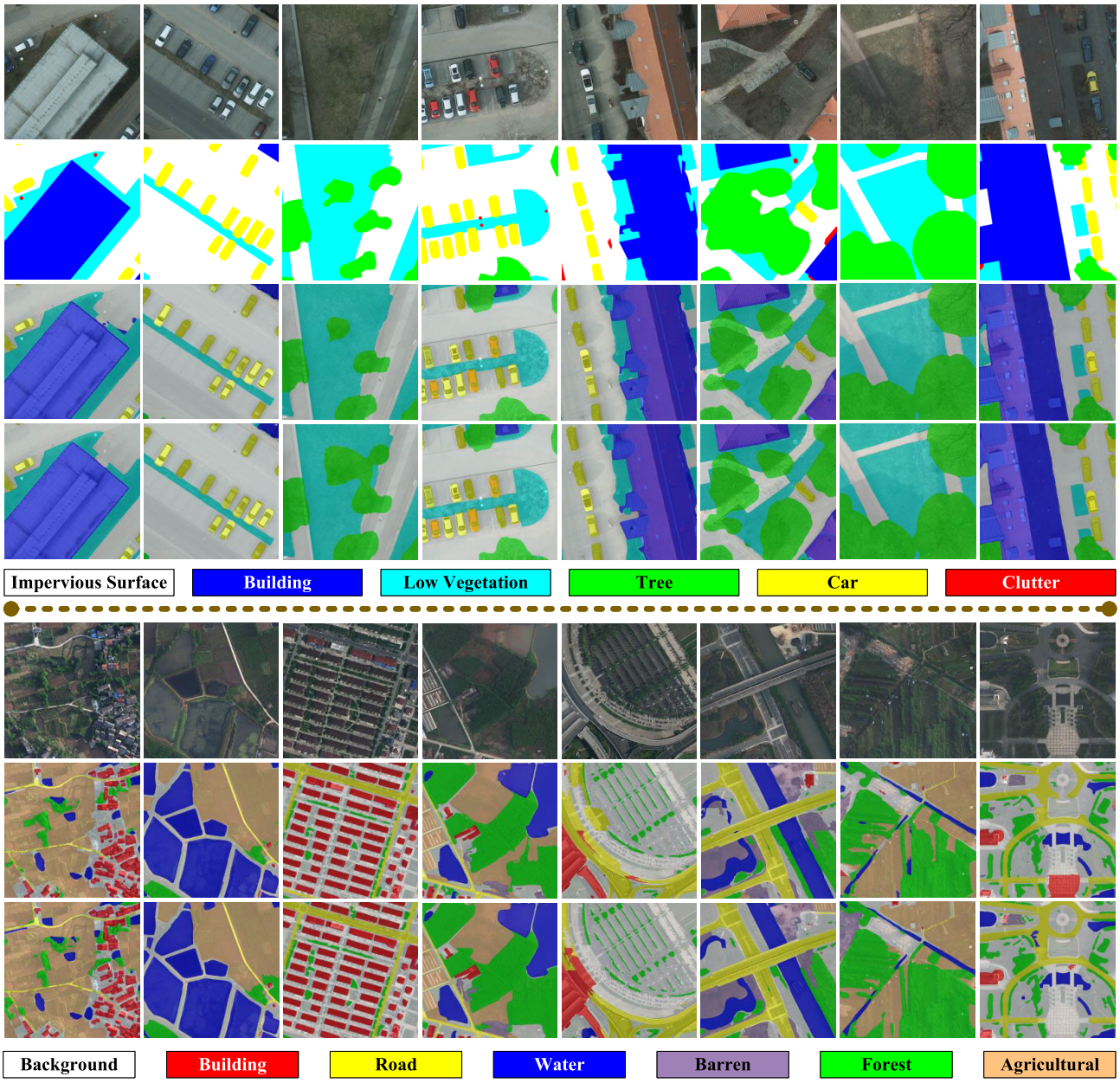


Fig. 7. Visualization results of the proposed model on semantic segmentation task. The first four rows are images, ground truth, predictions made by SMLFR-ConvNeXt-B and SMLFR-ConvNeXt-L of the Potsdam dataset, and the last three rows are images, predictions made by SMLFR-ConvNeXt-B and SMLFR-ConvNeXt-L of the LoveDA dataset.

configurations for the optimizer and the dynamic learning rate adjustment strategy are maintained consistently with those used for semantic segmentation. For improved localization accuracy of change regions in remote sensing imagery, we use the cross-entropy loss, measuring the discrepancy between predicted change regions and true change regions.

3) Experimental Results: The comparative experimental results for remote sensing change detection on the LEVIR-CD dataset are presented in Table VI. Table VI highlights the best (**red**) and suboptimal (**blue**) performance in each metric. It can be observed that the proposed models significantly outperform the competing techniques across three key metrics. Notably, with respect to the F1 score metric, SMLFR-ConvNeXt-B and SMLFR-ConvNeXt-L exhibit a substantial lead, surpassing the previous best-performing method RingMo [44] by 1.80% and 2.03%, respectively. It is worth noting that while SMLFR-ConvNeXt-L possesses more parameters and a larger model scale, its lead is not pronounced. This may be attributed to the limited sample size in the LEVIR-CD dataset, potentially resulting in model overfitting.

The visualization comparison on the LEVIR-CD dataset is displayed in Fig. 8. Our models exhibit excellent change detection capabilities and can effectively handle additional disturbances caused by seasonal differences or changes in the appearance of surface features. By deeply learning the effective contextual information of remote sensing image pairs, our models more accurately capture the real semantic changes and exclude irrelevant change factors. Ultimately, our method can generate relatively complete prediction results under a

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TABLE VI
COMPARATIVE RESULTS OF PRECISION, RECALL, AND F1 SCORE ON THE TEST SET OF THE LEVIR-CD DATASET

Method	Precision (%)	Recall (%)	F1 (%)
FC-EF [98]	86.91	80.17	83.40
FC-Siam-Conc [98]	91.99	76.77	83.69
FC-Siam-Di [98]	89.53	83.31	86.31
CLNet [99]	89.80	90.30	90.00
SNUNet-CD [100]	89.18	87.17	88.16
IFN [101]	94.02	82.93	88.13
DTCDSCN [102]	88.53	86.83	87.67
DASNet [103]	80.76	79.53	79.91
STANet [97]	83.81	91.00	87.34
BSFNet [104]	82.70	**94.00**	88.00
HRTNet [105]	85.43	91.77	88.48
CDNet [106]	91.60	86.50	89.00
BIT-RES [107]	89.24	89.37	89.31
ChangeFormer [108]	92.05	88.80	90.40
CS-HSNet [109]	93.12	88.58	90.79
BiT(Seco-ResNet-50) [87]	-	-	90.14
BiT(RSP-ViTAEv2-S) [88]	-	-	90.93
BiT(ISP) [44]	92.38	89.98	91.16
BiT(ISSP) [44]	92.04	89.76	90.94
BiT(RingMo) [44]	92.65	91.08	91.86
BiT(SMLFR-ConvNeXt-B)	**95.43**	92.03	**93.66**
BiT(SMLFR-ConvNeXt-L)	**95.02**	**92.83**	**93.89**

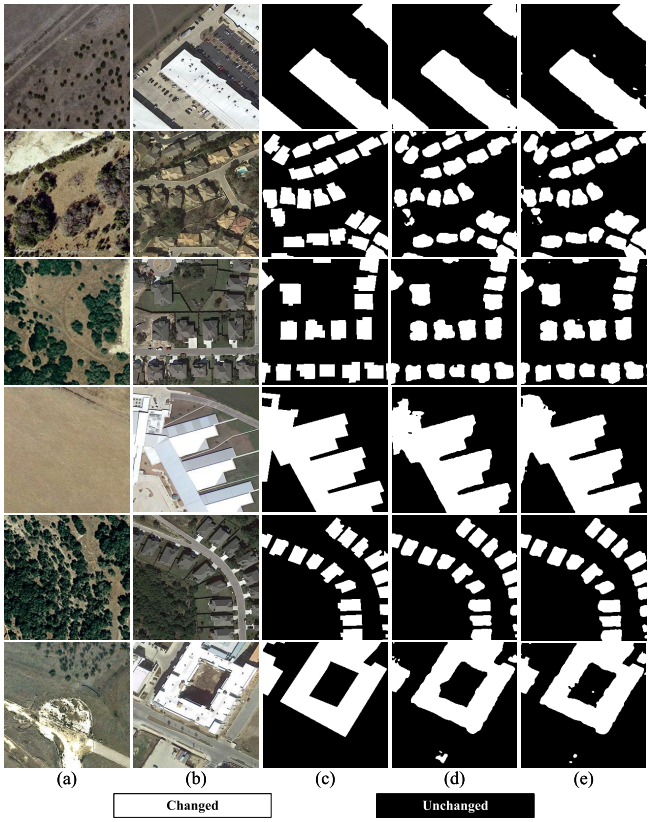


Fig. 8. Visualization results of different methods on the LEVIR-CD test set. (a) T0 images. (b) T1 images. (c) Ground truth. (d) SMLFR-ConvNeXt-B. (e) SMLFR-ConvNeXt-L.

wide range of changing scenarios, providing strong support for remote sensing data analysis.

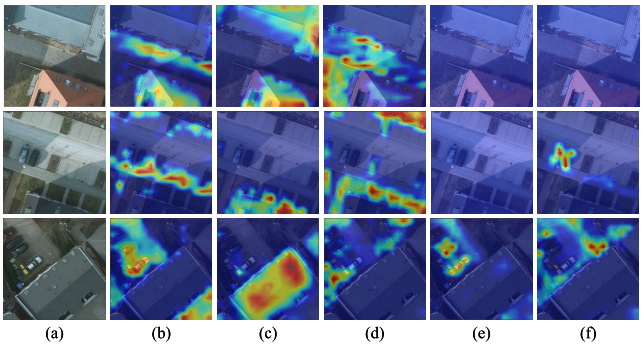


Fig. 9. Category-specific attention visualization in semantic segmentation on the Potsdam dataset. (a) Input image. (b)–(f) Attention heatmaps of SMLFR-ConvNeXt-L for categories impervious surface, building, low vegetation, tree, and car, respectively.

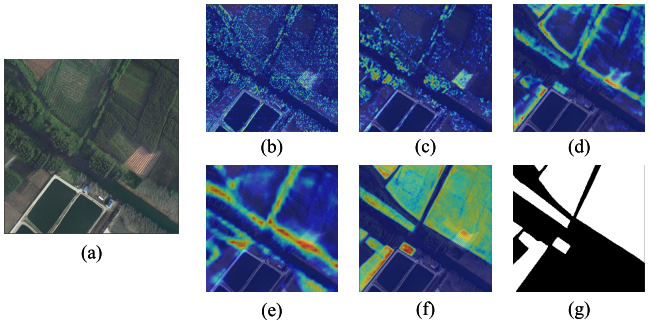


Fig. 10. Attention visualization for the agriculture category at different depths of SMLFR-ConvNeXt-B on the LoveDA Dataset. (a) Input image. (b)–(e) Attention maps of the last convolution layer in different stages of the encoder. (f) Attention map of the last normalization layer of the decoder. (g) Prediction result of SMLFR-ConvNeXt-B.

V. DISCUSSION

In this section, we use visualization tools to conduct a detailed exploration of the learned representations by our proposed foundation model.

In Fig. 9, we visualize the attention maps corresponding to each category in the Potsdam semantic segmentation dataset, which are generated by the last convolution layer of our proposed SMLFR-ConvNeXt-L model. These visualizations effectively demonstrate our model's exceptional contextual modeling and semantic understanding capabilities. Besides, the model also adeptly captures fine details, such as edges and intricate structures, in remote sensing images.

Furthermore, we present attention visualizations for the category Agriculture at different stages of the SMLFR-ConvNeXt-B model on the LoveDA Dataset, which can be seen in Fig. 10. In the initial stages of the encoder, the model primarily extracts low-level representations, encompassing textures and boundaries. As we progress toward the later stages of the encoder, the model demonstrates a pronounced advancement in semantic comprehension and discriminative capabilities, evident in its ability to extract higher level features. Besides, the attention maps of the decoder reveal the model's proficiency in synthesizing refined semantic information. This stage effectively integrates the context learned during encoding, resulting in an enhanced ability to discriminate and interpret the intricate details associated with the category Agriculture.

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VI. Conclusion

In this article, we propose a generative ConvNet foundation model for interpreting remote sensing images. Our proposed foundational model comprises two pivotal components: an extensive dataset called GeoSense and an SMLFR self-supervised learning framework. The GeoSense dataset is constructed to bolster the model's robustness and generalization during pretraining, which boasts a comprehensive collection of approximately nine million diverse remote sensing images. By incorporating sparse modeling and low-frequency reconstruction, the SMLFR is designed to guide the model's attention toward crucial ground object features in remote sensing images while mitigating interference from superfluous details. Extensive experiments were conducted across five datasets encompassing three downstream tasks, and the results consistently demonstrate that our proposed foundation model achieves state-of-the-art performance across all the benchmark datasets and downstream tasks, signifying its efficacy and potential for advancing remote sensing applications.

Despite the outstanding performance exhibited by current foundational vision models across various downstream tasks, the necessity for independent and comprehensive fine-tuning for each task significantly constrains the generalizability and transferability of the foundational models. In future work, we are committed to investigating parameter-efficient fine-tuning (PEFT) and continual learning to address the aforementioned limitations, with the aim of enhancing the usability of large-scale remote sensing foundational models in real-time scenarios and mobile systems.

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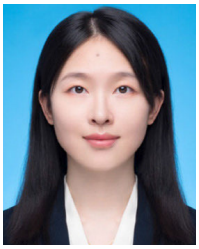
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